CR-002: Medicine Expiry Validation During Sales

1. Change Request Details

Field	Details
Change Request ID	CR-002
Title	Medicine Expiry Validation During Sales
Priority	High
Difficulty	Medium
Change Type	Functional Enhancement
Affected Version	Pharmacy Management System current baseline
Dependency	Existing sales workflow, medicine expiry dates and stock records

2. Business Requirement

The pharmacy must prevent the sale of expired medicines. Although expiry monitoring and sales-related expiry logic already exist, validation needs to be consistently enforced in both the user interface and the backend to reduce dispensing risk and protect stock accuracy.

3. Current Behaviour

• The system already includes medicine expiry monitoring.

• The sales route contains expiry-related logic, but stronger end-to-end validation is requested.

• A rejected or invalid sale must never cause an unintended stock deduction.

4. Requested Change

Strengthen medicine expiry validation throughout the sales process. The implementation must:

• Prevent expired medicines from being sold.

• Define and enforce a clear policy for medicines with missing expiry dates; the proposed policy is to block sale until a valid expiry date is recorded.

• Display an understandable warning when an expired medicine is selected.

• Validate expiry dates again on the backend, regardless of frontend checks.

• Reject invalid sale requests with a clear error response.

• Keep stock quantities unchanged when a sale is rejected.

5. Scope

In Scope

• Sales page selection and warning messages.

• Backend validation of expiry date and missing expiry date.

• Sale rejection and protection of stock quantities.

• Validation tests for expired, missing-date and valid medicines.

• Regression testing of the existing sales workflow.

Out of Scope

• New expiry notification scheduling or email/SMS alerts.

• Changes to supplier purchasing and reorder workflows.

• New database tables or schema migrations unless implementation review establishes a necessity.

• Changes to login, roles or unrelated modules.

6. Expected / Potentially Affected Files

• routes/sales.js

• public/js/sales.js

• public/pages/sales.html

• public/css/sales.css

Additional files, if modified, must be documented and justified in the Implementation Report.

7. Functional & Security Requirements

• FR-001: The sales interface shall identify and warn about expired selected medicines.

• FR-002: Expired medicines shall not be submitted as successful sales.

• FR-003: Medicines with a missing or invalid expiry date shall be blocked until corrected.

• FR-004: The backend shall independently validate expiry dates for every sale item.

• FR-005: Invalid sale requests shall return a clear validation error.

• FR-006: Rejected sales shall not reduce stock or create a completed sale.

• FR-007: Valid, unexpired medicines shall remain saleable subject to existing rules.

• SEC-001: Existing sales authentication and authorization shall remain enforced.

• SEC-002: Client-side validation shall not replace backend validation.

8. Acceptance Criteria

• AC-001: Selecting an expired medicine displays a visible warning.

• AC-002: A sale containing an expired medicine is rejected by the backend.

• AC-003: Missing or invalid expiry dates result in rejection under the defined policy.

• AC-004: Invalid requests return a meaningful error and do not complete the sale.

• AC-005: Stock remains unchanged after any rejected sale.

• AC-006: Valid unexpired medicines can be sold through the existing workflow.

• AC-007: Existing sales permissions and unrelated functionality continue working.

9. Expected Outcome

Expired or unverified-expiry medicines cannot be sold, users receive clear warnings, and rejected sales do not affect stock quantities.

10. Implementation Report Requirement

- Implementation summary and affected files.
- Validation policy and backend/frontend changes.
- Test cases, actual results and acceptance-criteria verification.
- Evidence screenshots, final status and outstanding issues.